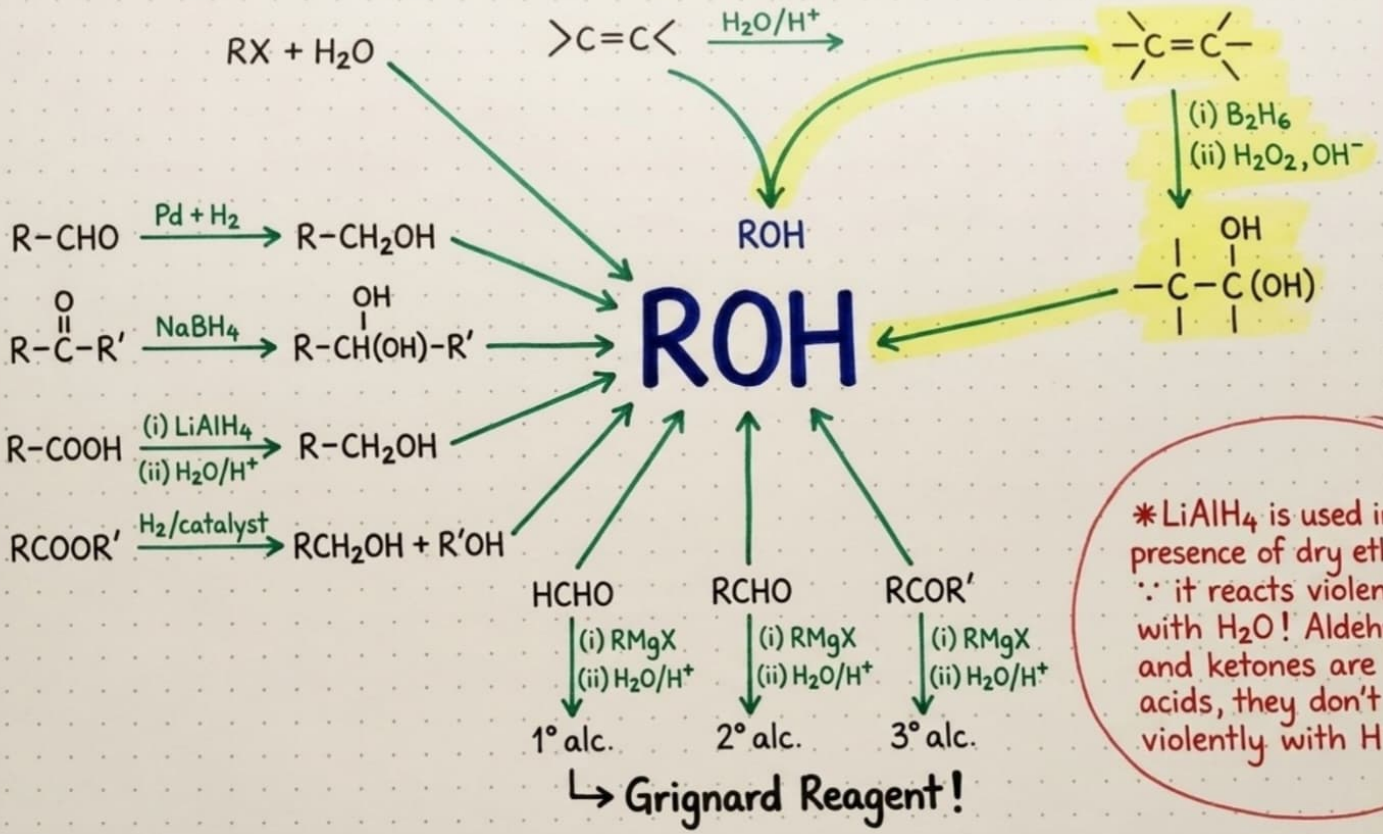
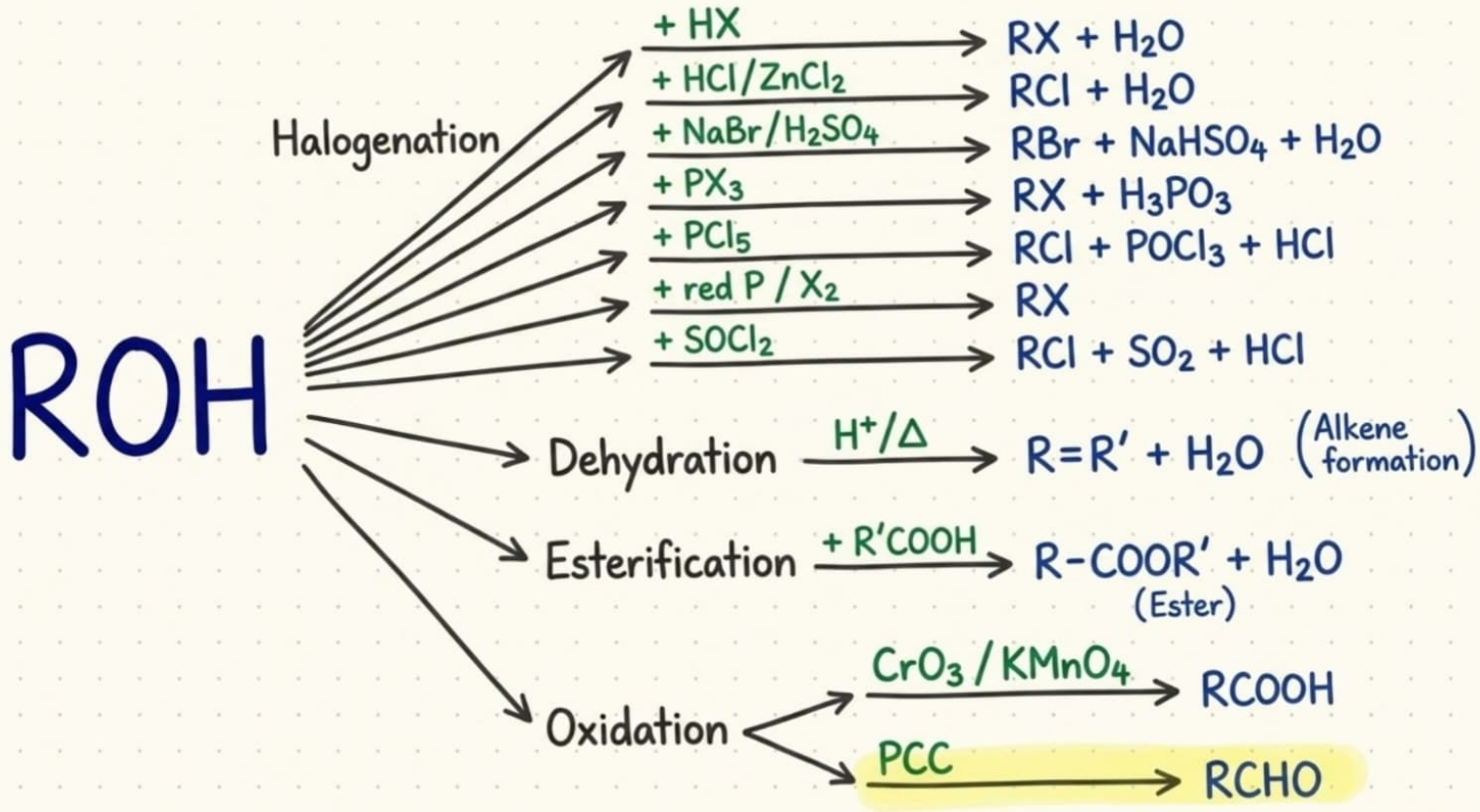
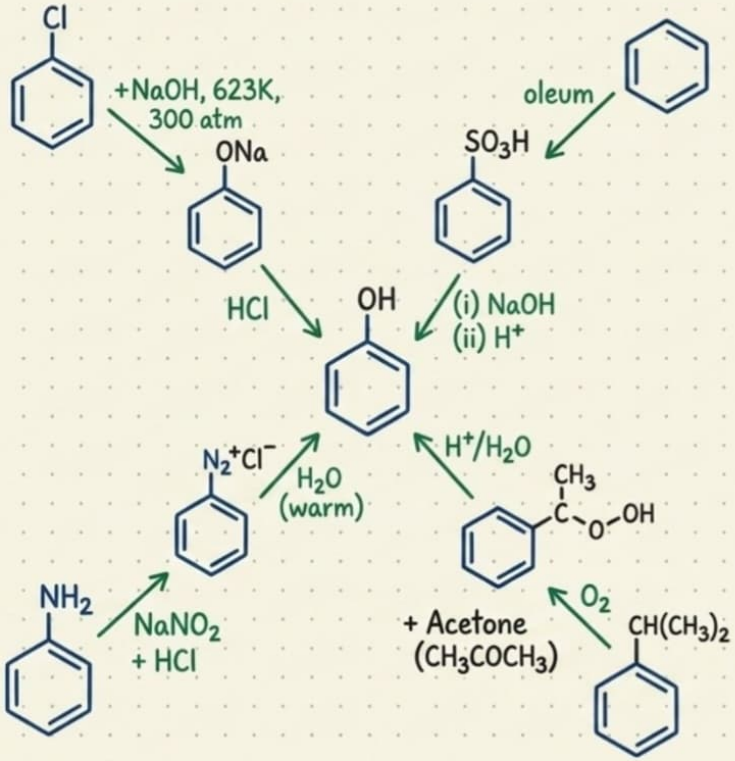
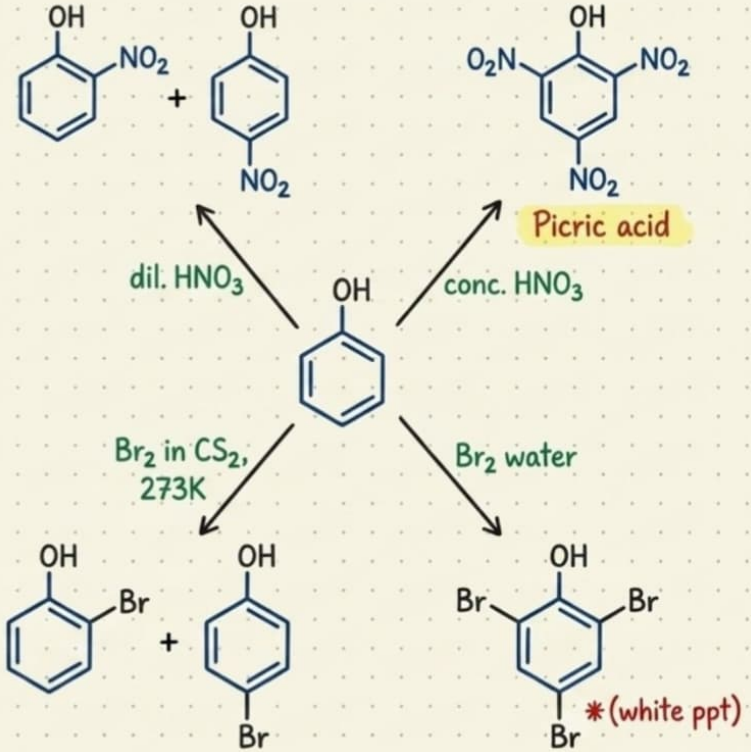

RX + H2O
>C=C<
H2O/H+
–C=C–
(i) B2H6
(ii) H2O2, OH−
OH
–C–C(OH)
ROH
R–CHO
Pd + H2
R–CH2OH
O
R–C–R′
NaBH4
OH
R–CH(OH)–R′
ROH
R–COOH
(i) LiAlH4
(ii) H2O/H+
R–CH2OH
RCOOR′
H2/catalyst
RCH2OH + R′OH
HCHO
RCHO
RCOR′
(i) RMgX
(ii) H2O/H+
1° alc.
2° alc.
3° alc.
↳ Grignard Reagent!
LiAlH4 is used in the presence of dry ether/THF ∵ it reacts violently with H2O! Aldehydes and ketones are poor acids, they don't react violently with H−


ROH
Halogenation
+ HX
RX + H2O
+ HCl/ZnCl2
RCl + H2O
+ NaBr/H2SO4
RBr + NaHSO4 + H2O
+ PX3
RX + H3PO3
+ PCl5
RCl + POCl3 + HCl
+ red P / X2
RX
+ SOCl2
RCl + SO2 + HCl
Dehydration
H+/Δ
R=R' + H2O
(Alkene formation)
Esterification
+ R'COOH
R-COOR' + H2O
(Ester)
Oxidation
CrO3 / KMnO4
RCOOH
PCC
RCHO

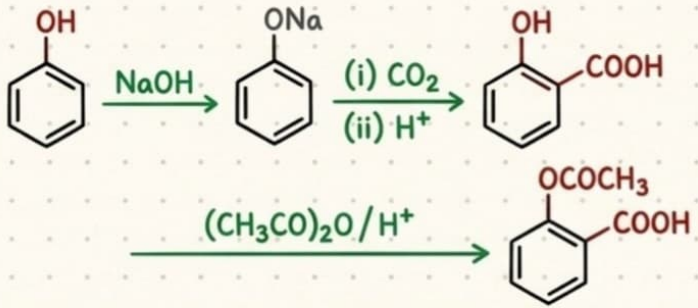
Preparation of Phenol



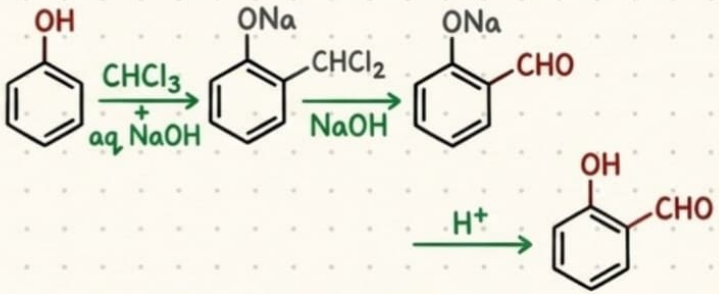
Basic Reactions



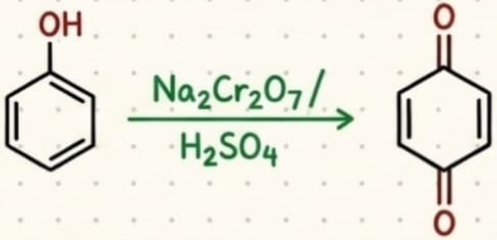
Kolbe Reaction



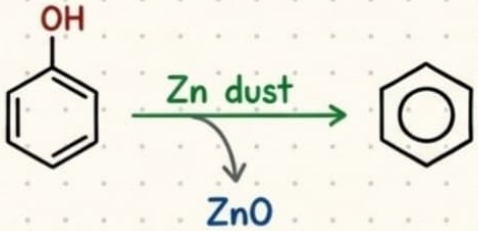
Reimer-Tiemann Reaction



Oxidation to Benzoquinone



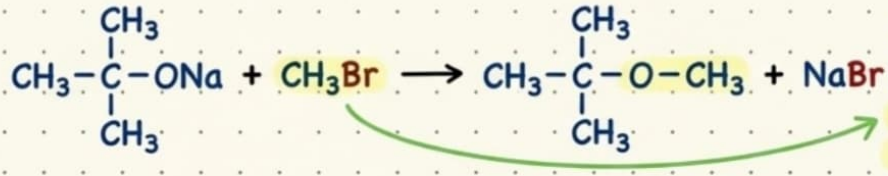
Zinc Dust Reduction



Williamson Ether Synthesis

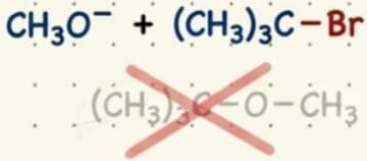
$$RX + R'\text{–}ONa \longrightarrow R\text{–}O\text{–}R' + NaX$$

The Optimal Path ✓



Smaller alkyl halide makes it easier to attack. Methyl halide is most suitable!

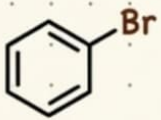
The Blocked Path ✗



⚠ A 3° halide cannot be attacked! Steric hindrance & +I effect of $-CH_3$ blocks substitution.

Elimination takes place instead.

The Impossible Path ✗



+ Nucleophile

Formation of ether is impossible. We can't break the Aryl–Halogen bond due to partial double bond character!

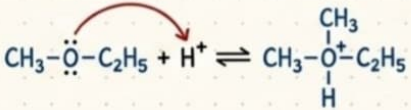
Cleavage by Hydrogen Halides

Reactivity order : HI > HBr > HCl

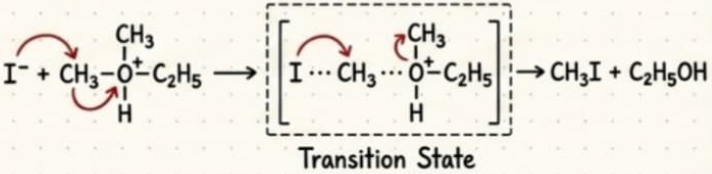
Cleavage takes place under drastic conditions with high temp.

The HI Cleavage Mechanism

Step I

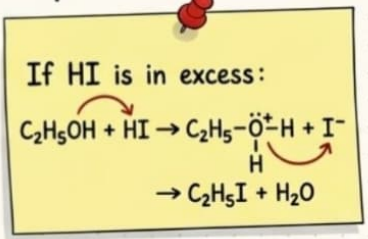
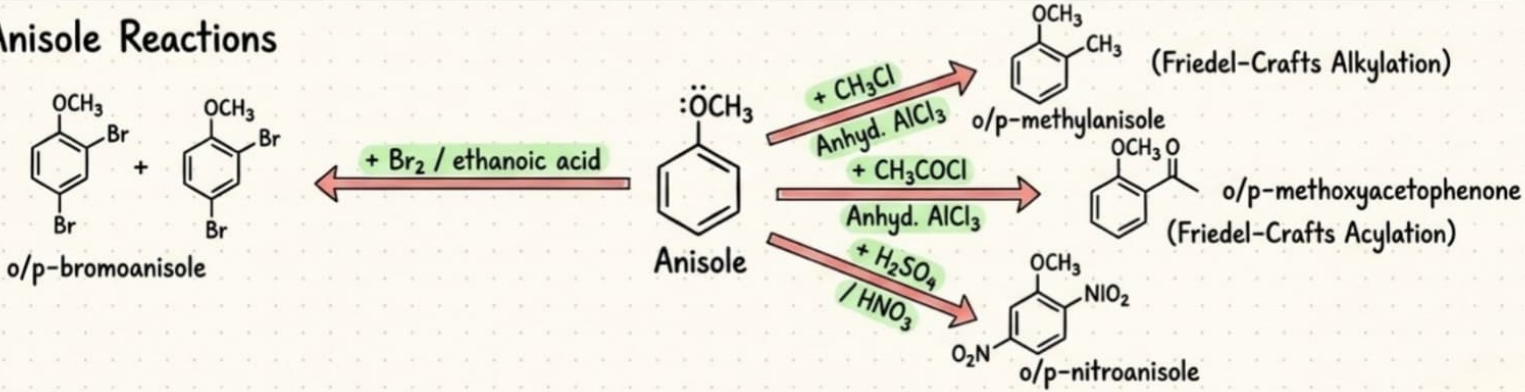


Step II



Step III

If HI is in excess:



Anisole Reactions